Solid Geometry & Similarity

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

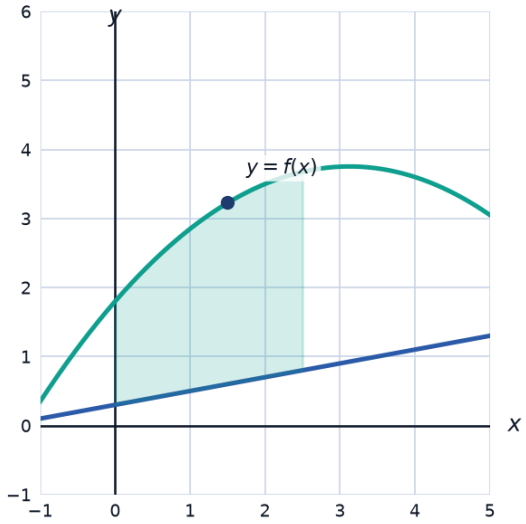
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Volume of prisms and cylinders

Find the volume of a cylinder with radius 5 cm and height 14 cm, in terms of π.

2 [Foundational] • Target: Volume of prisms and cylinders

A triangular prism has a triangular cross-section of area 24 cm^2 and length 15 cm. Find its volume.



3 [Foundational] • Target: Volume of prisms and cylinders

Find the volume of a cube with side length 9.

4 [Foundational] • Target: Volume of pyramids and cones

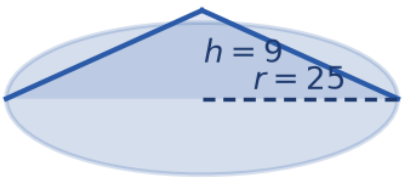
Find the volume of a square pyramid with base side 10 and height 12.

5 [Foundational] • Target: Volume of pyramids and cones

Find the volume of a cone with radius 6 and height 15, in terms of π.

6 [Foundational] • Target: Volume of pyramids and cones

A cone has volume 300π cubic units and height 25. Find its radius.



© ClipSAT — clipsat.org | All Rights Reserved

7 [Foundational] • Target: Volume of pyramids and cones

A rectangular pyramid has a base 6 by 8 and height 9. Find its volume.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Volume of spheres

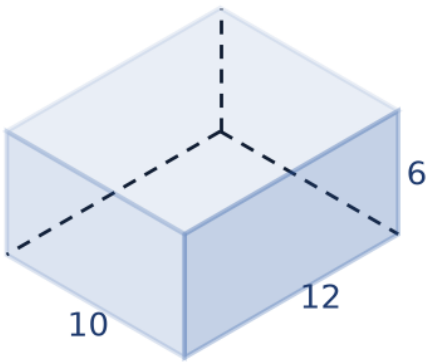
Find the volume of a sphere with radius 9, in terms of π.

9 [Intermediate] • Target: Volume of spheres

A hemisphere has radius 6. Find its volume, in terms of π.

10 [Intermediate] • Target: Volume of spheres

A spherical tank has diameter 10. Find its volume, in terms of π.



11 [Intermediate] • Target: Volume of spheres

Find the volume and surface area of a sphere with radius 3, in terms of π.

12 [Intermediate] • Target: Surface area of solids

Find the surface area of a cylinder with radius 4 and height 10, in terms of π.

13 [Intermediate] • Target: Surface area of solids

Find the surface area of a cone with radius 5 and slant height 13, in terms of π.

14 [Intermediate] • Target: Surface area of solids

Find the surface area of a sphere with radius 7, in terms of π.

© ClipSAT — clipsat.org | All Rights Reserved

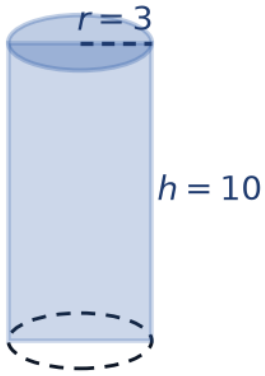
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Surface area of solids • Cross-Lesson: 8.01 Statistics & Probability

Find the surface area of a cube with side length 7.

16 [Challenging] • Target: Composite solids • Cross-Lesson: 1.01 Advanced Algebra

A solid consists of a cylinder (radius 3, height 10) topped by a cone with the same radius and height 4. Find the total volume, in terms of π.



17 [Challenging] • Target: Composite solids • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

A rectangular box measuring 8 by 6 by 4 has a cylindrical hole (radius 2, drilled fully through the height of 4) removed. Find the remaining volume, in terms of π.

18 [Challenging] • Target: Composite solids • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

A solid consists of a cylinder (radius 4, height 8) topped by a hemisphere of the same radius. Find the total volume, in terms of π.

19 [Challenging] • Target: Composite solids • Cross-Lesson: 4.01 Lines, Triangles & Polygons

A solid is a rectangular box (10 by 6 by 5) with a square pyramid (base 6 by 6, height 4) carved out of the top. Find the remaining volume.

20 [Challenging] • Target: Similar solids • Cross-Lesson: 5.01 Trigonometry

Two similar cylinders have heights 6 and 9. Find the ratio of their volumes.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Volume of prisms and cylinders

Two similar solids have volume ratio $1:64$. If a side of the smaller solid is 11, find the corresponding side of the larger solid.

a. 27

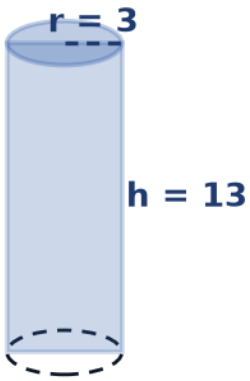
b. 19

c. 44

d. 30

22 [Foundational] • Target: Volume of pyramids and cones

A solid is formed by a cylinder of radius 3 and height 13 topped with a hemisphere of the same radius. Find the total volume. (In terms of π.)



a. 164π

b. 27π

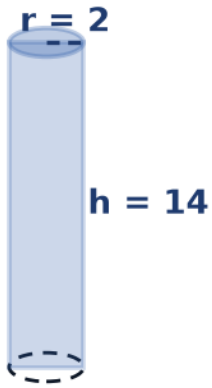
c. 135π

d. 51π

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Volume of spheres

Find the volume of a cylinder with radius 2 and height 14. (Leave your answer in terms of π.)



a. 78π

b. 61π

c. 56π

d. 22π

24 [Foundational] • Target: Surface area of solids

A cone with height H and base radius R is inscribed within a sphere of radius 5. The cone's vertex is at the sphere's south pole, and its base is a horizontal circle on the sphere. If the height of the cone is H = 8, what is the volume of the cone?

a. $\frac{128\pi}{3}$

b. $\frac{256\pi}{3}$

c. 128π

d. 256π

25 [Foundational] • Target: Composite solids

A cone has a height of 18 cm. A smaller cone is formed by slicing the top of the original cone with a plane parallel to the base. The smaller cone is similar to the original cone, and its height is 12 cm. If the radius of the base of the original cone is 9 cm, what is the radius of the base of the smaller cone?

a. 6 cm

b. 5 cm

c. 4.5 cm

d. 3 cm

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Similar solids

A right circular cone has a height of 12 cm. A smaller cone is formed by slicing the top of the larger cone with a plane parallel to the base. The smaller cone is similar to the larger cone, and its height is 6 cm. If the radius of the base of the larger cone is 9 cm, what is the radius of the base of the smaller cone?

a. 4.5 cm

b. 5 cm

c. 6 cm

d. 7.5 cm

27 [Foundational] • Target: Cross-sections of solids

In the diagram below, two similar triangles ABC and ADE are shown, with point D on segment AB and point E on segment AC. If AD = 4, DB = 8, and AE = 6, what is the length of EC?

a. 12

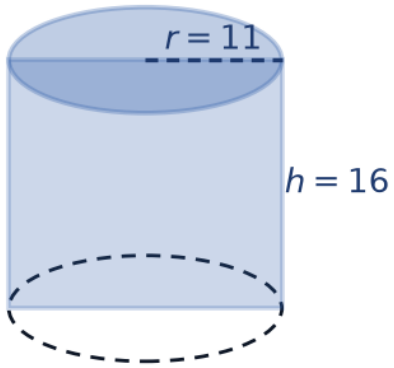
b. 10

c. 8

d. 9

28 [Foundational] • Target: A well-designed word problem: a grain silo

Find the volume of a cube with edge length 11.



a. 1331

b. 777

c. 814

d. 1001

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 [Intermediate] • Target: Volume of prisms and cylinders

A cylinder has volume 1053π and radius 9. Find its height.

a. 2

b. 13

c. 7

d. 15

30 [Intermediate] • Target: Volume of pyramids and cones

A solid cone has a height of 24 cm. A smaller cone is cut from the top of the larger cone by a plane parallel to the base. The smaller cone has a height of 8 cm. If the radius of the base of the larger cone is 12 cm, what is the radius of the base of the smaller cone?

a. 4 cm

b. 6 cm

c. 8 cm

d. 5 cm

31 [Intermediate] • Target: Volume of spheres

In the diagram below, chords AB and CD intersect at point P. If AP = 6, PB = 8, and CP = 4, find the length of PD.

a. 3

b. 4.5

c. 6

d. 12

32 [Intermediate] • Target: Surface area of solids

Find the angle θ between $\mathbf{u} = \langle 1, 1\sqrt{3} \rangle$ and $\mathbf{v} = \langle 0, 3 \rangle$.

a. 30°

b. 90°

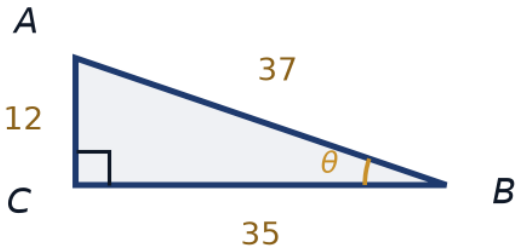
c. 60°

d. 45°

© ClipSAT — clipsat.org | All Rights Reserved

33 [Intermediate] • Target: Composite solids

Identify the Cartesian curve represented by $r = 6\cos\theta$.



a. Circle of radius 6 centered at $(0, 0)$

b. Circle of radius 3 centered at $(3, 0)$

c. Cardioid with parameter $a = 3$

d. Ellipse with semi-major axis 6

34 [Intermediate] • Target: Similar solids

Find the principal cube root of $8i$ in polar form.

a. $2(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})$

b. $8(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6})$

c. $2(\cos 0 + i\sin 0)$

d. $2(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6})$

35 [Intermediate] • Target: Cross-sections of solids

$\Delta ABC \sim \Delta DEF$ with side ratio $AB : DE = 2 : 3$. If the area of ΔABC is 24, find the area of ΔDEF.

a. 54

b. 53

c. 14

d. 63

36 [Intermediate] • Target: A well-designed word problem: a grain silo

Find the focus and directrix of the parabola $y^2 = 8x$.

a. Focus $(2, 0)$, Directrix $x = -2$

b. Focus $(-2, 0)$, Directrix $x = 2$

c. Focus $(8, 0)$, Directrix $x = -8$

d. Focus $(0, 2)$, Directrix $y = -2$

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: Volume of prisms and cylinders

Convert $r = \frac{4}{1 - 2\cos\theta}$ to Cartesian form and identify its type.

a. Parabola: $y^2 = 4(2)x$

b. Hyperbola: $3x^2 - y^2 + 16x + 16 = 0$

c. Circle: $x^2 + y^2 = 4$

d. Ellipse: $3x^2 + y^2 = 16$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Volume of pyramids and cones • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Convert polar coordinates $(r, \theta) = (4, \frac{\pi}{3})$ to rectangular (x, y).

a. $(2, -2\sqrt{3})$

b. $(-2, 2\sqrt{3})$

c. $(2\sqrt{3}, 2)$

d. $(2, 2\sqrt{3})$

39 [Challenging] • Target: Volume of spheres • Cross-Lesson: 4.01 Lines, Triangles & Polygons

A conic section has eccentricity $e = 0.20$. Identify this conic section.

a. Hyperbola

b. Circle

c. Parabola

d. Ellipse

40 [Challenging] • Target: Surface area of solids • Cross-Lesson: 5.01 Trigonometry

Find the equation of a sphere in $\mathbb{R}^3$ centered at $(1, -2, 2)$ with radius 3.

a. $x^2 + y^2 + z^2 = 9$

b. $(x - 1)^2 + (y + 2)^2 + (z - 2)^2 = 3$

c. $(x + 1)^2 + (y - 2)^2 + (z + 2)^2 = 9$

d. $(x - 1)^2 + (y + 2)^2 + (z - 2)^2 = 9$

© ClipSAT — clipsat.org | All Rights Reserved

41 [Challenging] • Target: Composite solids • Cross-Lesson: 6.01 Circles & Coordinate Geometry

An L-shaped region is formed by removing a 3×9 rectangular corner from a 20×20 rectangle. Find the area of the remaining region.

a. 373

b. 364

c. 382

d. 379

42 [Challenging] • Target: Similar solids • Cross-Lesson: 8.01 Statistics & Probability

Find the sum of the infinite series $8 + \frac{8}{2} + \frac{8}{4} + \cdots$.

a. 16

b. 20

c. Diverges

d. 8

43 [Challenging] • Target: Cross-sections of solids • Cross-Lesson: 1.01 Advanced Algebra

Two similar triangles have corresponding side lengths in ratio $4:6$. If the smaller triangle has area 8, what is the area of the larger triangle?

a. 10

b. 22

c. 14

d. 18

44 [Challenging] • Target: A well-designed word problem: a grain silo • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

A survey was conducted on the preferred modes of transportation for students travelling to school. The results are summarized in the two-way table below. If a student is chosen at random from those surveyed, and it is known that they prefer to travel by bus, what is the probability that they are in Year 10? Let N_{10} represent the number of Year 10 students who prefer the bus, and N_{Bus} represent the total number of students who prefer the bus. The probability we are looking for is $\frac{N_{10}}{N_{Bus}}$. | Year Group | Bicycle | Bus | Train | Walk | Total | |---|---|---|---|---|---| | Year 9 | 45 | 60 | 25 | 70 | 200 | | Year 10 | 55 | x | 35 | 80 | 270 | | Year 11 | 30 | 50 | 40 | 50 | 170 | | Total | 130 | 160 | 100 | 200 | 640 |

a. $\frac{130}{160}$

b. $\frac{x}{160}$

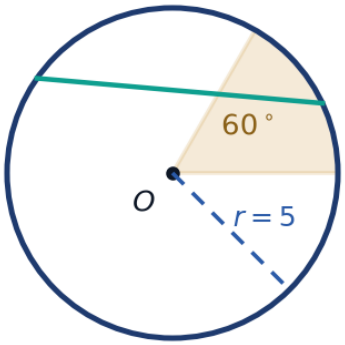
c. $\frac{x}{270}$

d. $\frac{160}{640}$

© ClipSAT — clipsat.org | All Rights Reserved

45 [Challenging] • Target: Volume of prisms and cylinders • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

A circle has a radius of 6 units. A chord within this circle subtends a sector with a central angle of 60 degrees. What is the length of the chord?



a. 6 units

b. $6\sqrt{3}$ units

c. 12 units

d. $12\sqrt{3}$ units

© ClipSAT — clipsat.org | All Rights Reserved